CS445 / SE463 / CS645
Software Requirements Specification and Analysis Spring 2026

Deliverable 5: Quality Requirements and Prioritizing Requirements

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July 13, 2026

1: Updated Behavioural Model

No changes were required from D4. Below is a copy of the state diagram from D4.

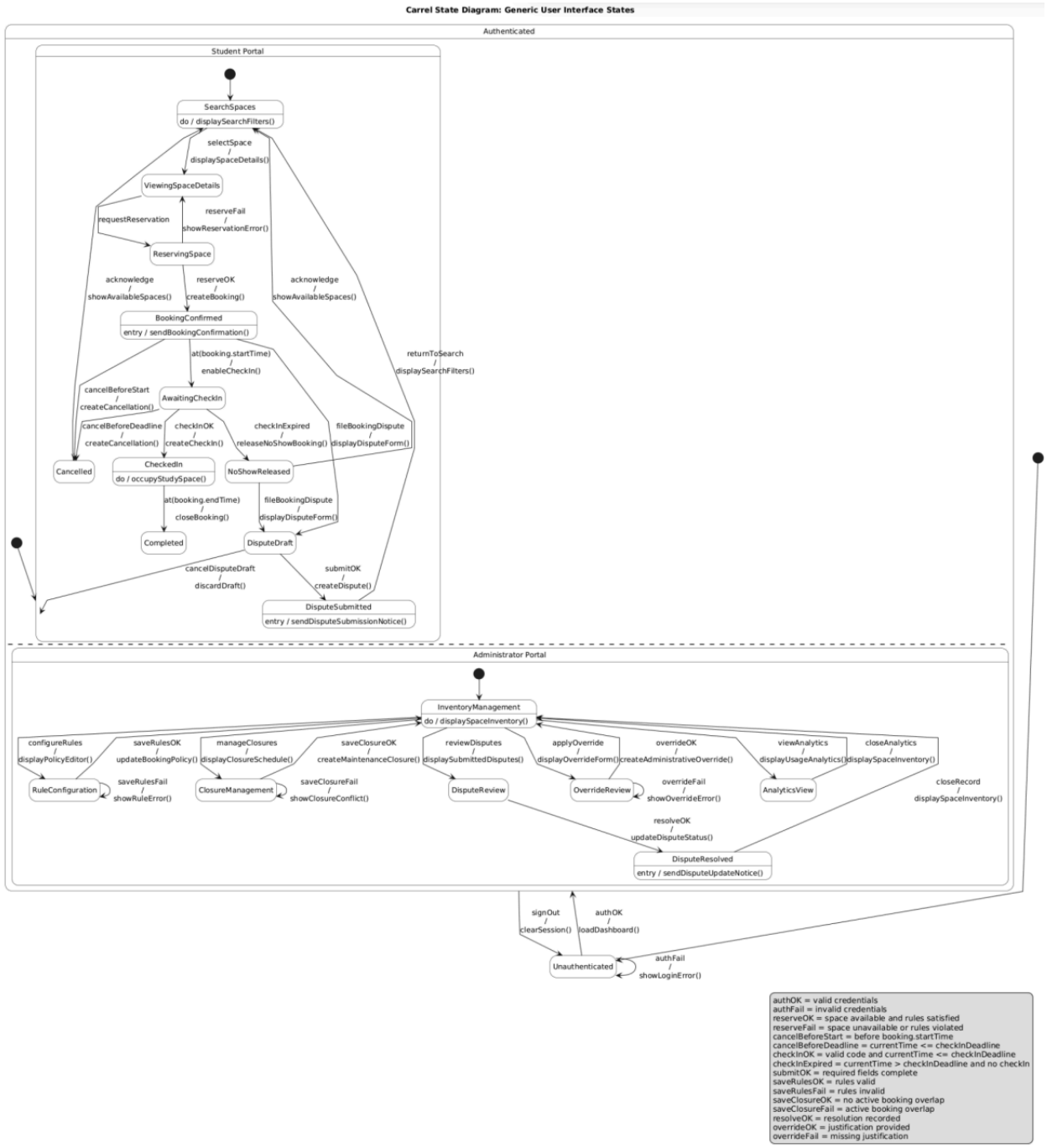


Figure 1: updated state diagram (unchanged from D4)

2. Quality requirements

2.1 Performance

Peak load refers to the beginning-of-term weekday load profile: 500 concurrently active user sessions, of which roughly 60 percent are searching or viewing spaces, 25 percent are creating or cancelling bookings, and 15 percent are checking in.

PER-1 (search response time). When a student performs an availability search (UC1), Carrel shall return the first page of matching study spaces within 2.0 seconds at the 95th percentile under peak load.

Measurement: a scripted load test (for example JMeter or k6) is run for 30 minutes against a production-equivalent staging deployment using the peak load profile, and the 95th percentile of server response time for search requests is computed.

Fit criterion	Outstanding	Target	Minimum
95th percentile search response time at peak load	1.0 s	2.0 s	4.0 s

PER-2 (booking transaction time). A booking request (UC3) shall complete within 3.0 seconds at the 95th percentile under peak load, measured from submission of the reservation form to the booking being persisted with status Confirmed and its check-in code issued.

Measurement: server-side timestamps recorded during the same 30-minute load test; the interval is measured from request receipt to commit of the Confirmed booking and creation of the CheckInCode record.

Fit criterion	Outstanding	Target	Minimum
95th percentile booking completion time at peak load	1.5 s	3.0 s	5.0 s

PER-3 (check-in throughput). Carrel shall sustain at least 20 QR check-in validations (UC5) per second for 5 consecutive minutes while keeping the 95th percentile validation latency at or

below 1.5 seconds, corresponding to the burst of students arriving at study spaces when classes end.

Measurement: a load generator replays valid check-in requests at a fixed arrival rate for 5 minutes against the staging deployment; the sustained rate and the 95th percentile of validation latency are recorded.

Fit criterion	Outstanding	Target	Minimum
Sustained check-in rate, with p95 latency bound	40/s at 1.0 s	20/s at 1.5 s	10/s at 3.0 s

2.2 Reliability

REL-1 (availability). Carrel shall be at least 99.5 percent available during the academic term, measured monthly over the daily service window 07:00 to 24:00 Eastern Time, excluding the planned maintenance window defined in OPC-3.

Measurement: an external uptime monitor probes the search endpoint every 60 seconds; monthly availability = successful probes / total probes within the service window.

Fit criterion	Outstanding	Target	Minimum
Monthly availability in the service window	99.9%	99.5%	99.0%

REL-2 (no-show release correctness rate). At least 99.9 percent of no-show releases (UC17) in a term shall execute successfully within 60 seconds of the booking's check-in deadline, so that unclaimed spaces reliably return to the available pool.

Measurement: at end of term, a database query compares releasedAt against checkInDeadline for every booking that reached status NoShowReleased, and counts bookings that were eligible for release but were released late or not at all.

Fit criterion	Outstanding	Target	Minimum
Releases executed successfully, within delay bound	99.99% within 30 s	99.9% within 60 s	99.5% within 5 min

REL-3 (mean time between failures). The mean time between failures of the booking service shall be at least 30 days, where a failure is any unplanned incident that prevents students from creating or cancelling bookings for more than 60 consecutive seconds.

Measurement: computed from the operations incident log over each four-month term as in-service hours divided by the number of qualifying incidents.

Fit criterion	Outstanding	Target	Minimum
MTBF of the booking service	90 days	30 days	14 days

2.3 Integrity

INT-1 (no conflicting bookings). Carrel shall never persist two bookings with status in {Confirmed, AwaitingCheckIn, CheckedIn} for the same study space with overlapping time intervals, even under concurrent booking attempts on the same space and time slot.

Measurement: a concurrency stress test submits 10,000 pairs of deliberately conflicting booking requests with randomized interleaving against the staging deployment, then a conflict-detection query searches for overlapping active bookings; the same query runs nightly against the production database as an audit.

Fit criterion	Outstanding	Target	Minimum
Overlapping active bookings persisted	0 in stress test and 0 in all nightly production audits over the term	0 in stress test; any production violation detected within 24 h, auto-corrected, and both students notified	at most 1 production violation per term, corrected within 24 h of detection

INT-2 (backup and verified recovery). Carrel shall back up all booking, policy, and dispute data at least every 24 hours; after each backup the system shall verify the backup copy against the live database (checksums and record counts) and report any discrepancy, so that at most 24 hours of data can be lost in a storage failure.

Measurement: the percentage of backups in a term that pass automated verification is computed from backup logs, and a timed restore drill into a clean environment is performed once per term.

Fit criterion	Outstanding	Target	Minimum
Verified backups; timed restore drill	100% pass; restore in 1 h; hourly incrementals (RPO 1 h)	99% pass; restore in 4 h; RPO 24 h	97% pass; restore in 24 h; RPO 24 h

INT-3 (lifecycle audit accuracy). Every booking lifecycle transition (confirmation, check-in, cancellation, no-show release, administrative override) shall be recorded in an append-only audit log with actor, timestamp, and prior state, and every student's currentStrikeCount shall equal the number of that student's active NoShowStrike records.

Measurement: an automated reconciliation job recomputes strike counts from NoShowStrike records for all students daily and reports discrepancies; system tests verify that each lifecycle operation writes exactly one audit record.

Fit criterion	Outstanding	Target	Minimum
Reconciliation discrepancies and audit completeness	0 discrepancies all term; 100% of transitions audited	no discrepancy unresolved longer than 48 h; 100% audited	discrepancy rate at most 0.1% of students per month, all corrected within 7 days

2.4 Robustness

ROB-1 (Calendar Service outage tolerance). If the Calendar Service is unavailable, booking (UC3) shall still complete successfully; calendar invitations (UC16) shall be queued and automatically delivered within 15 minutes of service recovery, and the confirmation shown to the student shall state that the invitation is delayed.

Measurement: a fault-injection test simulates a 60-minute Calendar Service outage while 100 bookings are performed; the number of failed bookings and the delay between service recovery and invitation delivery are recorded.

Fit criterion	Outstanding	Target	Minimum
Bookings failed during outage; invite delay after recovery	0 failed; delivered within 5 min	0 failed; delivered within 15 min	at most 2% of bookings complete without invitation, with user notice; delivered within 60 min

ROB-2 (invalid input tolerance). Carrel shall reject malformed or invalid requests (invalid time ranges, unknown spaces, expired or tampered check-in codes, empty dispute reasons) with a specific error message identifying the problem, shall write no partial data, and shall not crash.

Measurement: a fuzz test submits 10,000 generated malformed requests across the booking, check-in, cancellation, and dispute endpoints; crashes, unspecific error responses, and orphaned partial records are counted.

Fit criterion	Outstanding	Target	Minimum
Fuzz test outcome over 10,000 malformed requests	0 crashes; 100% specific messages; 0 partial writes	0 crashes; 95% specific messages; 0 partial writes	0 crashes; 90% specific messages; partial writes automatically rolled back

ROB-3 (identity service outage tolerance). If the University Identity Service is unavailable, sessions that are already authenticated shall continue to operate for at least 30 minutes, new login attempts shall receive a clear service-unavailable message within 5 seconds, and normal login shall resume within 2 minutes of the identity service recovering.

Measurement: a fault-injection test disables the identity service integration for 45 minutes while active sessions continue working; session survival time, the response to new login attempts, and time to recovery are recorded.

Fit criterion	Outstanding	Target	Minimum
Session survival / login error response / recovery time	60 min / 2 s / 1 min	30 min / 5 s / 2 min	15 min / 10 s / 5 min

2.5 Security

SEC-1 (authentication and role enforcement). Every Carrel function shall require authentication through the University Identity Service, and administrator functions (UC8 to UC13) shall be accessible only to accounts holding the Administrator role; no student or anonymous credential shall be able to invoke any administrator operation.

Measurement: a penetration test before each release exercises every administrator endpoint with student tokens, anonymous requests, and manipulated role claims, and records findings by severity.

Fit criterion	Outstanding	Target	Minimum
Penetration test findings per release	0 findings of any severity	0 high-severity findings; medium findings fixed within 7 days	0 high-severity findings; all other findings fixed within 30 days

SEC-2 (session expiry). Authenticated sessions shall expire after at most 30 minutes of inactivity and require re-authentication, and session tokens shall be invalidated server-side upon sign-out so that a captured token cannot be replayed.

Measurement: automated security tests hold a session idle past the threshold and then attempt an action (expecting rejection), and replay a token captured before sign-out (expecting rejection).

Fit criterion	Outstanding	Target	Minimum
Idle timeout enforced, sign-out replay rejected	15 min; replay rejected	30 min; replay rejected	60 min; replay rejected

SEC-3 (encryption in transit). All communication between clients and Carrel, and between Carrel and its partner services (University Identity Service, Calendar Service), shall be encrypted with TLS version 1.2 or higher, and any plain HTTP request shall be redirected to HTTPS.

Measurement: a quarterly automated TLS configuration scan (for example SSL Labs) of all public endpoints, plus a configuration audit of internal service connections.

Fit criterion	Outstanding	Target	Minimum
TLS scan grade and protocol floor	A+ with TLS 1.3 only	A with TLS 1.2 or higher	B with no cipher on the known-vulnerable list and remediation within 30 days

2.6 Process Requirements

PRC-1 (change review and standards compliance). All source code changes shall reach the main branch only through a pull request approved by at least one team member other than the author, with the continuous integration pipeline (build, automated tests, and linter with the project's coding standard) passing before merge.

Measurement: at each release, repository history and CI logs are analyzed to compute the percentage of merged pull requests that satisfied the review and pipeline conditions.

Fit criterion	Outstanding	Target	Minimum
Compliant merged pull requests	100%, with 2 approvals required on booking-core modules	100% with at least 1 approval	95%, with every exception documented

PRC-2 (automated test coverage). Automated tests shall cover at least 80 percent of statements in the booking-core modules (reservation, check-in, cancellation, and no-show release), enforced as a merge gate in continuous integration.

Measurement: the CI coverage report for the main branch at each release; the gate blocks any merge that would drop booking-core statement coverage below the threshold.

Fit criterion	Outstanding	Target	Minimum
Statement coverage of booking-core modules	90%	80%	70%

PRC-3 (delivery schedule). Carrel shall be delivered in staged milestones: a pilot covering the student booking flow (UC1 to UC6) shall be available to a test cohort by August 24, 2026, and the full system covering all 17 use cases shall be released by September 8, 2026, the first day of the fall term.

Measurement: release tags and deployment records are compared against the milestone dates in the project plan.

Fit criterion	Outstanding	Target	Minimum
Milestone dates	pilot 2 weeks early, full release on time	both dates met	pilot on time, full release at most 2 weeks late

2.7 Operating Constraints

Module 10 characterizes operating constraints by the location of the system, its size and power consumption, ambient conditions, operating costs, and accessibility for maintenance. The three constraints below instantiate the aspects applicable to a hosted campus web system: location, operating costs, and maintenance accessibility.

OPC-1 (hosting location and data residency). Carrel's production servers and all stored student data, including backups, shall be located in the university's on-campus data centre; any supplementary cloud capacity shall be located in a Canadian region approved by campus IT, so that student personal data never leaves Canadian jurisdiction.

Measurement: a deployment configuration audit at each release lists every component, data store, and backup location and checks it against the approved-location list; the required proportion in approved locations is 100 percent.

Fit criterion	Outstanding	Target	Minimum
Component locations	all components including backups on campus	campus primary with Canadian-region backup	Canadian-region cloud under a signed data-processing agreement

OPC-2 (operating cost). The total operating cost of running Carrel (compute, storage, monitoring, and licences) shall not exceed 500 Canadian dollars per month averaged over each academic term, so the service stays within the student-services budget envelope.

Measurement: hosting invoices and licence statements for the term are summed and divided by the number of months.

Fit criterion	Outstanding	Target	Minimum
Average monthly operating cost	CA$250	CA$500	CA$750

OPC-3 (maintenance window and accessibility). All planned maintenance shall occur within the Sunday 02:00 to 06:00 Eastern Time window, and operators shall be able to deploy a new release or roll back to the previous release within 30 minutes inside that window.

Measurement: the change-management log is reviewed each term for out-of-window maintenance events, and a timed deployment-and-rollback drill is performed once per term.

Fit criterion	Outstanding	Target	Minimum
Out-of-window events per term; rollback time	0 events; 10 min	at most 1 event with 48 h notice; 30 min	at most 2 events with 48 h notice; 60 min

3 Prioritizing Requirements: Cost-Value Approach

We applied the Cost-Value Approach from Module 12 (slides 10 to 19), which is based on the Analytic Hierarchy Process (AHP), to one requirement from each of the seven categories of Section 2. The chosen requirements are the architecturally most significant of their categories:

Label	Requirement	Category
PER-1	Search responds within 2.0 s (p95) at peak load	Performance
REL-1	99.5% availability during the term	Reliability
INT-1	No conflicting bookings ever persisted	Integrity
ROB-1	Bookings survive Calendar Service outages	Robustness
SEC-1	Identity-service authentication and role enforcement	Security
PRC-2	80% automated test coverage of booking-core modules	Process
OPC-1	On-campus hosting and Canadian data residency	Operating constraint

Following the module, the approach has five steps:
(I) review the candidate requirements for completeness and clarity;
(II) determine the relative value of the requirements using AHP pairwise comparison;
(III) repeat the AHP estimation for relative cost;
(IV) draw the cost-value diagram;
(V) use the diagram to analyze and discuss the requirements. Within Steps II and III, the AHP computation itself has five sub-steps (M12 slide 13), and the consistency check has five sub-steps (M12 slide 15); we show all of them for both value and cost.

All calculations below were carried out in a spreadsheet-style script at full precision and are displayed rounded (matrix entries to 3 decimals, weights to 4 decimals, percentages to 1 decimal), so displayed rows may differ from displayed totals by 0.001 and percentages may not sum to exactly 100.0. The formulas used are stated with each sub-step.

3.1 Step I: Reviewing the candidate requirements

The requirements engineers (all three group members) reviewed the seven candidate requirements to ensure they are complete and clearly defined. Each candidate is fully specified in Section 2 with a measurable statement, a measurement procedure, and a richer fit criterion, so each pairwise judgment below compares two well-defined, testable requirements rather than vague goals.

3.2 Step II: Relative value by AHP pairwise comparison

Process. The three group members made each judgment together, in the role of Carrel's stakeholders (students and space administrators), asking: which of these two requirements, if satisfied at its Target level, contributes more to stakeholder satisfaction, and how strongly? Judgments use the AHP scale from M12 slide 12: 1 means the requirements are of equal value, 3 means one is slightly preferred, 5 strongly preferred, 7 very strongly preferred, 9 highly preferred, with even values 2, 4, 6, 8 used as compromises between adjacent judgments. We filled the upper triangle of the matrix (21 comparisons for n = 7); each cell (x, y) below the diagonal automatically receives the reciprocal 1/n of its mirror cell, and the diagonal is 1. Table 1 documents every judgment and its reasoning.

Table 1: Pairwise value judgments and reasoning (the number states how many times more valuable the first requirement is than the second)

Pair	Judgment	Reasoning
PER-1 vs REL-1	1	Fast search and high availability both shape the everyday student experience; neither dominates the other.
PER-1 vs INT-1	1/2	Conflict-free bookings matter more than speed: a fast system that double-books is useless, so INT-1 is moderately more valuable.
PER-1 vs ROB-1	1	Search speed affects every session mildly; calendar-outage tolerance affects rare sessions more severely. These balance out.
PER-1 vs SEC-1	1/2	Preventing unauthorized access protects all users and the university; it outweighs snappy search.

PER-1 vs PRC-2	3	Students directly feel search speed; test coverage benefits them only indirectly through fewer defects.
PER-1 vs OPC-1	2	Visible everyday speed is worth somewhat more to stakeholders than hosting location, though residency has privacy-compliance value.
REL-1 vs INT-1	1/2	Downtime is a recoverable annoyance; conflicting booking data breaks the core promise and causes real-world room disputes.
REL-1 vs ROB-1	2	Total unavailability blocks all functions; a Calendar Service outage only delays a convenience feature.
REL-1 vs SEC-1	1	Uptime and security are both foundational trust properties of a campus service; equal value.
REL-1 vs PRC-2	4	Users feel downtime immediately and personally; coverage is a developer-facing safeguard.
REL-1 vs OPC-1	2	Being up matters more day to day than where the servers sit.
INT-1 vs ROB-1	3	The no-double-booking guarantee is the product's core fairness promise; degraded calendar behaviour is peripheral.
INT-1 vs SEC-1	1	Both protect trust in the booking data: accidental corruption and unauthorized modification are equally damaging.
INT-1 vs PRC-2	7	Conflict-free data is the single most valuable property to stakeholders; coverage contributes only indirectly.
INT-1 vs OPC-1	4	Core data correctness clearly outweighs hosting location.
ROB-1 vs SEC-1	1/2	Security protects every user's account and data; calendar degradation touches one convenience feature.
ROB-1 vs PRC-2	3	Graceful degradation is visible to users; process metrics are not.
ROB-1 vs OPC-1	1	Both are secondary properties: one continuity of a convenience, one jurisdictional compliance; equal value.

SEC-1 vs PRC-2	5	A breach causes direct, possibly irreversible harm; coverage is an indirect quality lever, so SEC-1 is strongly preferred.
SEC-1 vs OPC-1	3	Access control is the broader trust property; residency is one narrower privacy control
PRC-2 vs OPC-1	1/2	Residency has direct compliance value to the university; coverage's value is indirect.

Sub-step 1 (compare pairs of requirements). The complete comparison matrix assembled from Table 1:

	PER-1	REL-1	INT-1	ROB-1	SEC-1	PRC-2	OPC-1
PER-1	1	1	1/2	1	1/2	3	2
REL-1	1	1	1/2	2	1	4	2
INT-1	2	2	1	3	1	7	4
ROB-1	1	1/2	1/3	1	1/2	3	1
SEC-1	2	1	1	2	1	5	3
PRC-2	1/3	1/4	1/7	1/3	1/5	1	1/2
OPC-1	1/2	1/2	1/4	1	1/3	2	1

Sub-step 2 (normalize the columns). Each entry is divided by the sum of its column (formula: normalized(i,j) = a(i,j) / column_sum(j)). The column sums are:

	PER-1	REL-1	INT-1	ROB-1	SEC-1	PRC-2	OPC-1
Column sum	7.833	6.250	3.726	10.333	4.533	25.000	13.500

The normalized matrix:

	PER-1	REL-1	INT-1	ROB-1	SEC-1	PRC-2	OPC-1
PER-1	0.128	0.160	0.134	0.097	0.110	0.120	0.148
REL-1	0.128	0.160	0.134	0.194	0.221	0.160	0.148
INT-1	0.255	0.320	0.268	0.290	0.221	0.280	0.296
ROB-1	0.128	0.080	0.089	0.097	0.110	0.120	0.074
SEC-1	0.255	0.160	0.268	0.194	0.221	0.200	0.222
PRC-2	0.043	0.040	0.038	0.032	0.044	0.040	0.037
OPC-1	0.064	0.080	0.067	0.097	0.074	0.080	0.074

Sub-steps 3, 4, 5 (sum each row; normalize sums by dividing by n = 7; report relative values as percentages):

Requirement	Sub-step 3: row sum	Sub-step 4: row sum / 7	Sub-step 5: relative value
PER-1	0.897	0.1282	12.8%
REL-1	1.144	0.1634	16.3%
INT-1	1.931	0.2758	27.6%
ROB-1	0.698	0.0998	10.0%
SEC-1	1.520	0.2171	21.7%
PRC-2	0.274	0.0392	3.9%
OPC-1	0.535	0.0765	7.6%

Checking consistency (M12 slide 15). Sub-step 1: multiply the comparison matrix by the priority vector (formula: (Aw)(i) = sum over j of a(i,j) × w(j)). Sub-step 2: divide each element of the result by the corresponding element of the priority vector. Both are shown per requirement:

Requirement	Priority w	Sub-step 1: A·w	Sub-step 2: (A·w)/w
PER-1	0.1282	0.908	7.088
REL-1	0.1634	1.156	7.072
INT-1	0.2758	1.956	7.090
ROB-1	0.0998	0.704	7.059
SEC-1	0.2171	1.538	7.081
PRC-2	0.0392	0.277	7.071
OPC-1	0.0765	0.542	7.084

Sub-step 3: the principal eigenvalue is the average of the sub-step 2 column:
$\lambda_{max} = (7.088 + 7.072 + 7.090 + 7.059 + 7.081 + 7.071 + 7.084) / 7 = 7.0778$

Sub-step 4: the consistency index is $CI = \frac{\lambda_{max} - n}{n-1} = \frac{7.0778 - 7}{6} = 0.0130.$

Sub-step 5: comparing against the consistency index of a random matrix: for $n = 7$, the random index is $RI = 1.32$ (M12 slide 16), so $CR = \frac{CI}{RI} = \frac{0.0130}{1.32} = 0.0098$. Since $0.0098 < 0.10$, the value judgments are consistent and the relative values can be used.

3.3 Step III: Relative cost by AHP pairwise comparison

Process. We repeated the identical procedure for relative cost. Here the question for each pair was: which requirement costs more to implement to its Target fit-criterion level, and how strongly? Cost includes development effort, required infrastructure, and the ongoing operational effort implied by the measurement procedure. The same 1 to 9 scale and reciprocal rule apply. Table 2 documents every judgment.

Table 2: Pairwise cost judgments and reasoning (the number states how many times more costly the first requirement is than the second)

Pair	Judgment	Reasoning
PER-1 vs REL-1	1	Both demand production-grade infrastructure work: caching, indexing, and a load-testing rig versus redundancy, health checks, and monitoring; comparable cost.
PER-1 vs INT-1	2	Meeting p95 targets under 500-session load requires stack-wide tuning and load infrastructure; the conflict guarantee is mostly careful transactional design plus tests.
PER-1 vs ROB-1	2	Performance work spans the whole stack; the invitation queue with retry is one bounded subsystem.
PER-1 vs SEC-1	2	Identity-service SSO and role checks reuse existing campus infrastructure; performance engineering is broader and more specialized.
PER-1 vs PRC-2	2	Reaching 80% coverage on core modules is significant but routine effort; performance tuning needs more specialized work.
PER-1 vs OPC-1	3	Campus hosting is mostly provisioning and configuration, clearly cheaper than performance engineering.
REL-1 vs INT-1	2	99.5% availability needs redundant instances, failover, and operational practices; conflict prevention is design and testing effort.
REL-1 vs ROB-1	2	Availability work covers every component; queueing covers one integration.
REL-1 vs SEC-1	3	High-availability infrastructure is the most expensive line item; SSO reuse is cheap by comparison.
REL-1 vs PRC-2	3	Infrastructure plus ongoing operations outweighs test-writing effort.
REL-1 vs OPC-1	3	Redundant infrastructure versus data-centre provisioning and configuration.
INT-1 vs ROB-1	1	Transactional locking with audit queries versus an outbox queue with fault-injection tests: similar bounded efforts.

INT-1 vs SEC-1	1	Both are moderate design-plus-testing efforts.
INT-1 vs PRC-2	1	Both consist mainly of disciplined engineering and automated tests.
INT-1 vs OPC-1	2	Stress tests and nightly audits still cost more than location configuration.
ROB-1 vs SEC-1	1	A bounded queueing subsystem versus SSO integration: similar.
ROB-1 vs PRC-2	1	Similar moderate efforts.
ROB-1 vs OPC-1	1	Both modest efforts.
SEC-1 vs PRC-2	1	Similar moderate efforts.
SEC-1 vs OPC-1	1	Both lean on existing campus infrastructure.
PRC-2 vs OPC-1	1	Both modest efforts.

Sub-step 1 (compare pairs of requirements). The cost comparison matrix:

	PER-1	REL-1	INT-1	ROB-1	SEC-1	PRC-2	OPC-1
PER-1	1	1	2	2	2	2	3
REL-1	1	1	2	2	3	3	3
INT-1	1/2	1/2	1	1	1	1	2
ROB-1	1/2	1/2	1	1	1	1	1
SEC-1	1/2	1/3	1	1	1	1	1
PRC-2	1/2	1/3	1	1	1	1	1
OPC-1	1/3	1/3	1/2	1	1	1	1

Sub-step 2 (normalize the columns). Column sums:

	PER-1	REL-1	INT-1	ROB-1	SEC-1	PRC-2	OPC-1
Column sum	4.333	4.000	8.500	9.000	10.000	10.000	12.000

The normalized matrix (each entry divided by its column sum):

	PER-1	REL-1	INT-1	ROB-1	SEC-1	PRC-2	OPC-1
PER-1	0.231	0.250	0.235	0.222	0.200	0.200	0.250
REL-1	0.231	0.250	0.235	0.222	0.300	0.300	0.250
INT-1	0.115	0.125	0.118	0.111	0.100	0.100	0.167
ROB-1	0.115	0.125	0.118	0.111	0.100	0.100	0.083
SEC-1	0.115	0.083	0.118	0.111	0.100	0.100	0.083
PRC-2	0.115	0.083	0.118	0.111	0.100	0.100	0.083
OPC-1	0.077	0.083	0.059	0.111	0.100	0.100	0.083

Sub-steps 3, 4, 5 (sum each row; divide by n = 7; report relative costs):

Requirement	Sub-step 3: row sum	Sub-step 4: row sum / 7	Sub-step 5: relative cost
PER-1	1.588	0.2269	22.7%
REL-1	1.788	0.2555	25.5%
INT-1	0.836	0.1194	11.9%
ROB-1	0.752	0.1075	10.7%
SEC-1	0.711	0.1015	10.2%
PRC-2	0.711	0.1015	10.2%
OPC-1	0.614	0.0876	8.8%

Checking consistency. Sub-steps 1 and 2 (multiply comparison matrix by priority vector; divide each element by the corresponding priority element):

Requirement	Priority w	Sub-step 1: A·w	Sub-step 2: (A·w)/w
PER-1	0.2269	1.605	7.075
REL-1	0.2555	1.808	7.079
INT-1	0.1194	0.846	7.089
ROB-1	0.1075	0.759	7.059
SEC-1	0.1015	0.716	7.053
PRC-2	0.1015	0.716	7.053
OPC-1	0.0876	0.619	7.059

Sub-step 3:
$\lambda_{max} = (7.075 + 7.079 + 7.089 + 7.059 + 7.053 + 7.053 + 7.059) / 7 = 7.0668$

Sub-step 4: $CI = \frac{7.0668-7}{6} = 0.0111$.

Sub-step 5: $CR = \frac{CI}{RI} = \frac{0.0111}{1.32} = 0.0084 < 0.10$, so the cost judgments are consistent.

3.4 Step IV: Cost-value diagram

Each requirement is plotted at the point (relative cost, relative value) from Steps II and III, with value on the y-axis and cost on the x-axis, both in percent. Following the module, the two lines through the origin partition the plane into priority regions: above the line value = 2 × cost a requirement gives a high value-to-cost ratio (high priority); below the line value = 0.5 × cost it gives a low ratio (low priority); between the lines the ratio is medium.

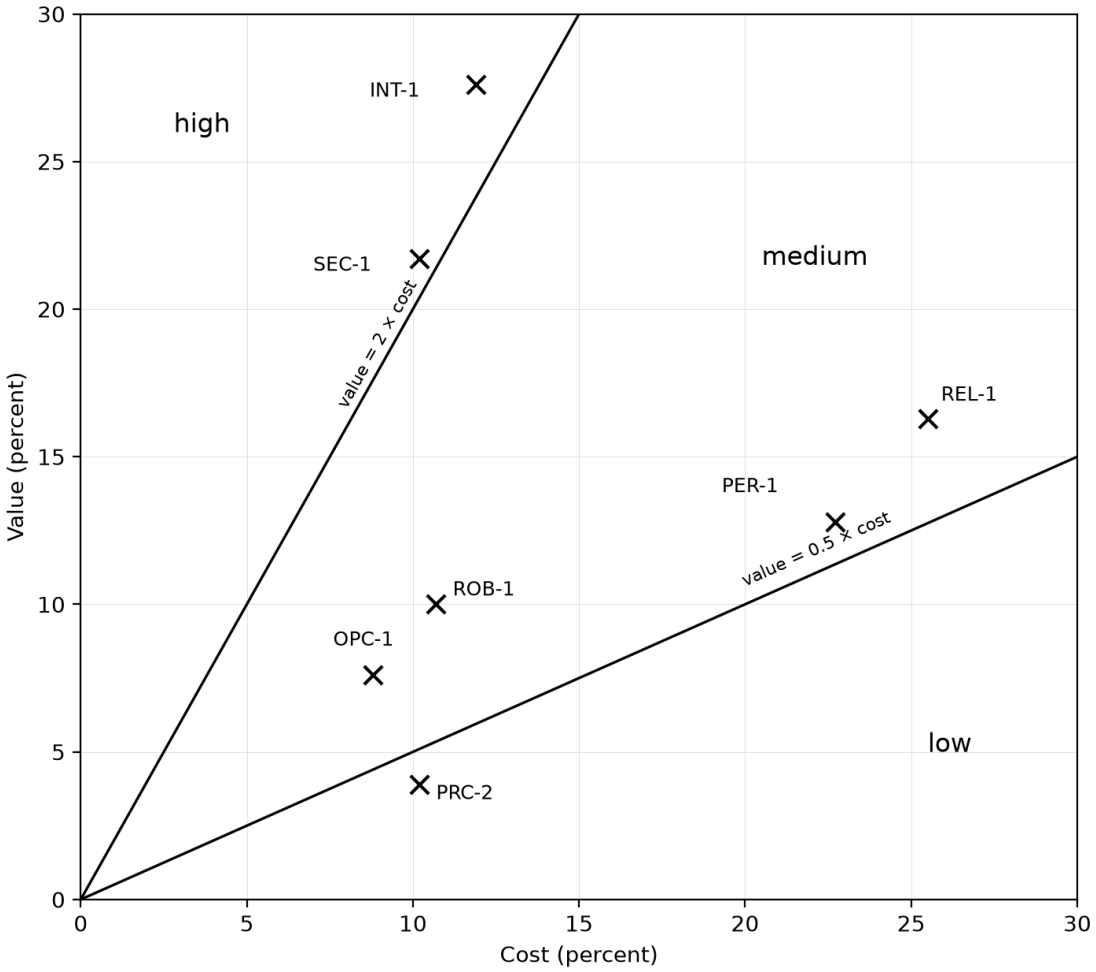


Figure 2: Carrel cost-value diagram

Requirement	Relative value	Relative cost	Value/cost ratio	Region
PER-1	12.8%	22.7%	0.56	Medium
REL-1	16.3%	25.5%	0.64	Medium
INT-1	27.6%	11.9%	2.31	High
ROB-1	10.0%	10.7%	0.93	Medium
SEC-1	21.7%	10.2%	2.14	High
PRC-2	3.9%	10.2%	0.39	Low
OPC-1	7.6%	8.8%	0.87	Medium

3.5 Step V: Using the diagram

The stakeholders use the diagram as a conceptual map for analyzing and discussing the requirements, and it yields a clear implementation strategy for Carrel. INT-1 (no conflicting bookings) and SEC-1 (authentication and role enforcement) fall in the high-priority region: together they carry about half of the total stakeholder value (49.3 percent) for barely more than a fifth of the total cost (22.1 percent), so they should be designed in from the first release and never traded away. REL-1 (availability) and PER-1 (search response time) carry substantial value but are the two most expensive requirements, which places them in the medium region; if budget or schedule pressure arises, these are the requirements whose Target levels should be renegotiated first (for example, relaxing availability toward the Minimum fit criterion of 99.0 percent) rather than dropping them. PER-1 sits just above the low-priority boundary, which reflects that its cost is driven by the peak-load condition; a cheaper interim tactic is to meet the response-time target at reduced load first. ROB-1 and OPC-1 are medium-priority requirements with nearly balanced value and cost, appropriate for the second iteration. PRC-2 (test coverage) lands in the low-priority region on this map; however, the cost-value diagram measures direct stakeholder value only, and the group still retains PRC-2 because it is a process safeguard that protects INT-1 and REL-1 over the system's lifetime, at the reduced Minimum threshold if effort must be cut. This matches the module's guidance that the diagram is a basis for discussion rather than an automatic decision rule.